Toy-car investigation: data collection

STUDENT WORKSHEET

Name: _____ Class: _____ Date: _____

Set-up and measurements

Investigation: How does the position of a toy car change with time on a shallow, fixed incline?

1. Draw and label the ramp and marking system. Show x = 0 and the positive direction (if not already provided).

2. Record the fixed ramp set-up and describe how you release the car. Use the same set-up for all five runs.

Position, x (m)	Time taken for toy car to reach position x from the origin (s)					
	t_1	t_2	t_3	t_4	t_5	Mean t
0.00						
0.05						
0.10						
0.15						
0.20						

Analysis on A4 graph paper

3. Plot x (vertical axis) against mean t (horizontal axis); both in SI units. Use a large sensible scale.

4. Draw a smooth curve of best fit that represents the trend, rather than joining each point with jagged lines.

5. Draw a tangent at the time when x = 0.05 m and another at the time when x = 0.15 m. Show two points used on each tangent and calculate the slopes, with units.

6. Compare the two estimates of velocity. What does this tell you about the car's motion?

Keep your original results for Lesson 5.